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SPECIALIST MATHEMATICS TRIAL EXAMINATION 1 SOLUTIONS 2025

Question 1 (3 marks)

Let $P(z) = z^3 - 2z^2 + 3z + c = 0$.

Given $c \in R$, all coefficients are real, therefore using the conjugate root theorem, $z = 2 - yi$ is also a solution.

(1 mark) – recognise CRT solution

We can rewrite $P(z)$ as the product of its factors:

$P(z) = (z - 2 + yi)(z - 2 - yi)(z + a)$, where $a \in R$ given third factor/solution must be real (or else it would need its conjugate pair).

$$\begin{aligned}\therefore P(z) &= ((z-2)^2 - (yi)^2)(z+a)\\ &= (z^2 - 4z + 4 + y^2)(z+a)\\ &= z^3 + (a-4)z^2 + (4 + y^2 - 4a)z + a(4 + y^2)\end{aligned}$$

Equating coefficients for z^2 term:

$$\begin{aligned}a - 4 &= -2\\ a &= 2\end{aligned}$$

(1 mark)

Equating the coefficients for the z term:

$$\begin{aligned}4 + y^2 - 4a &= 3\\ 4 + y^2 - 8 &= 3\\ y^2 &= 7\\ y &= \sqrt{7} \qquad \text{(given } y > 0\text{)}\end{aligned}$$

Equating the constant terms:

$$\begin{aligned}a(4 + y^2) &= c\\ 2(4 + 7) &= c\\ c &= 22\end{aligned}$$

(1 mark) – both c and y correct

Question 2 (4 marks)

a. Method 1

The distance from a line in vector equation form to a point is given by $D = \dfrac{|\overrightarrow{AP} \times \underset{\sim}{d}|}{|\underset{\sim}{d}|}$,

where A is the point given, P is any point on the line, e.g. $(2, 2, -1)$ and $\underset{\sim}{d}$ is a vector in the direction of the line, i.e. $\underset{\sim}{d} = 2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}$.

$$\overrightarrow{AP} = -2\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}$$

$$\therefore D = \frac{|(-2\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}) \times (2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k})|}{|2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}|} \qquad \textbf{(1 mark)}$$

$$= \frac{|-5\underset{\sim}{i} + 2\underset{\sim}{j} - 4\underset{\sim}{k}|}{|2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}|}$$

$$= \frac{\sqrt{(-5)^2 + 2^2 + (-4)^2}}{\sqrt{2^2 + 1^2 + (-2)^2}}$$

$$= \frac{\sqrt{45}}{3} = \frac{3\sqrt{5}}{5} = \sqrt{5} \qquad \textbf{(1 mark)}$$

Method 2

Let P be any point on the line and given the point A:

$$\overrightarrow{AP} = (2 + 2t)\underset{\sim}{i} + (2 + t)\underset{\sim}{j} - (1 + 2t)\underset{\sim}{k} - 4\underset{\sim}{i} - \underset{\sim}{j} + 4\underset{\sim}{k}$$
$$= (-2 + 2t)\underset{\sim}{i} + (1 + t)\underset{\sim}{j} + (3 - 2t)\underset{\sim}{k}$$

and let $\underset{\sim}{d} = 2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}$.

$$\therefore \overrightarrow{AP} \cdot \underset{\sim}{d} = 2(-2 + 2t) + 1 + t - 2(3 - 2t)$$
$$= -4 + 4t + 1 + t - 6 + 4t$$
$$= 9t - 9 \qquad \textbf{(1 mark)}$$

The closest point on the line to the point A is when the dot product found above is equal to 0.

$$\therefore 9t - 9 = 0$$
$$t = 1$$

Substitute $t = 1$ back into $\overrightarrow{AP}$:

$$\overrightarrow{AP} = 2\underset{\sim}{j} + \underset{\sim}{k}$$
$$\left|\overrightarrow{AP}\right| = \sqrt{2^2 + 1^2} = \sqrt{5} \qquad \textbf{(1 mark)}$$

b. Given the lines $\overrightarrow{AP} = -2\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}$ and $\underset{\sim}{d} = 2\underset{\sim}{i} + \underset{\sim}{j} - 2\underset{\sim}{k}$ are in the plane, we use their cross product to find the vector perpendicular to both which is a normal to the plane.

In part **a.** we found the cross product to be $-5\underset{\sim}{i} + 2\underset{\sim}{j} - 4\underset{\sim}{k}$, therefore the vector equation of the plane is given by:

$$-5x + 2y - 4z = k \qquad , \ k \in R$$

(1 mark)

Substitute a point on the plane into the equation to find k, e.g. $(2, 2, -1)$.

$$-5 \times 2 + 2 \times 2 - 4 \times -1 = k$$
$$k = -2$$

Therefore the equation of the plane required is $-5x + 2y - 4z = -2$.

(1 mark)

(Alternatively, the equation of the plane could be $5x - 2y + 4z = 2$).

Question 3 (4 marks)

a. According to the formula sheet, a confidence interval is found using $\left(\bar{x} - z\frac{s}{\sqrt{n}}, \bar{x} + z\frac{s}{\sqrt{n}}\right)$, where $\bar{x}$ is the sample mean, s is the standard deviation, n is the sample size and z is the value associated with the standard normal random variable. For a 95% confidence interval, we use $z = 1.96$ as given in the question.

$$\begin{aligned} \text{95\% confidence interval} &= \left(41 - 1.96 \times \frac{4.5}{\sqrt{81}}, 41 + 1.96 \times \frac{4.5}{\sqrt{81}}\right) && \textbf{(1 mark)} \\ &= (41 - 1.96 \times 0.5, 41 + 1.96 \times 0.5) \\ &= (41 - 0.98, 41 + 0.98) \\ &= (40.02, 41.98) && \textbf{(1 mark)} \end{aligned}$$

b. For the sample mean to differ from the population mean by 0.5 or less, we need $z\frac{s}{\sqrt{n}} = 0.5$, as a maximum.

$$\begin{aligned} \therefore z \times \frac{4.5}{\sqrt{81}} &= 0.5 \\ z \times 0.5 &= 0.5 \\ z &= 1 \end{aligned}$$

(1 mark)

Given $\Pr(Z > 1) = 0.16$, $\Pr(Z < -1) = 0.16$ and therefore $\Pr(-1 < Z < 1) = 1 - 2 \times 0.16 = 0.68$.

Therefore, we have a 68% confidence interval for the sample mean differing by 0.5 or less from the population mean.

$x = 68$. **(1 mark)**

Note: 0.68 as a solution is not an acceptable answer to the question due to the way the question is worded.

Question 4 (4 marks)

$$a = v\frac{dv}{dx} = \sqrt{v^2 - 4}$$
$$\frac{dv}{dx} = \frac{\sqrt{v^2 - 4}}{v}$$
$$\frac{dx}{dv} = \frac{v}{\sqrt{v^2 - 4}}$$
$$x = \int \frac{v}{\sqrt{v^2 - 4}}\, dv$$ **(1 mark)**

Let $u = v^2 - 4$

$$\therefore \frac{du}{dx} = 2v$$
$$\therefore x = \int \frac{1}{2} \times \frac{du}{dv} \times \frac{1}{\sqrt{u}}\, dv$$
$$= \frac{1}{2}\int \frac{1}{\sqrt{u}}\, du$$
$$= \sqrt{u} + c$$

$\therefore x = \sqrt{v^2 - 4} + c$ **(1 mark)**

When $x = 2$, $v = 6$, $\therefore 2 = \sqrt{32} + c$

$c = 2 - \sqrt{32} = 2 - 4\sqrt{2}$

$$x = \sqrt{v^2 - 4} + 2 - 4\sqrt{2}$$
$$\sqrt{v^2 - 4} = x + 4\sqrt{2} - 2$$

$\therefore v = \pm\sqrt{(x + 4\sqrt{2} - 2)^2 + 4}$ **(1 mark)**

Given $v = 6$ when $x = 2$, check which solution works.

$\therefore v = \sqrt{(x + 4\sqrt{2} - 2)^2 + 4}$ **(1 mark) –** pick correct solution

Question 5 (3 marks)

a. Using partial fractions:

$$\frac{1}{n(n-2)} = \frac{A}{n} + \frac{B}{n-2}$$
$$= \frac{A(n-2)}{n(n-2)} + \frac{Bn}{n(n-2)}$$
$$\therefore 1 = A(n-2) + Bn$$

Let $n = 0$:

$$1 = -2A$$
$$A = -\frac{1}{2}$$

Let $n = 2$:

$$1 = 2B$$
$$B = \frac{1}{2}$$

$$\therefore \frac{1}{n(n-2)} = -\frac{1}{2n} + \frac{1}{2(n-2)}$$
$$= \frac{1}{2}\left(\frac{1}{n-2} - \frac{1}{n}\right) \qquad \text{as required}$$

(1 mark) – sufficient working for show that

b. left hand side = $\frac{1}{3\times1} + \frac{1}{5\times3} + \frac{1}{7\times5} + \ldots.. + \frac{1}{n(n-2)}$

$$= \frac{1}{2}\left(\frac{1}{1} - \frac{1}{3}\right) + \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right) + \frac{1}{2}\left(\frac{1}{5} - \frac{1}{7}\right) + \ldots.. + \frac{1}{2}\left(\frac{1}{n-2} - \frac{1}{n}\right)$$

using partial fractions from part **a.** **(1 mark)**

This is a telescoping series, whereby terms cancel out, leaving just the first part of the first bracket and the last part of the last bracket.

$$\frac{1}{2}\left(\frac{1}{1} - \frac{1}{3}\right) + \frac{1}{2}\left(\frac{1}{3} - \frac{1}{5}\right) + \frac{1}{2}\left(\frac{1}{5} - \frac{1}{7}\right) + \ldots.. + \frac{1}{2}\left(\frac{1}{n-2} - \frac{1}{n}\right)$$
$$= \frac{1}{2}\left(\frac{1}{1} - \cancel{\frac{1}{3}} + \cancel{\frac{1}{3}} - \cancel{\frac{1}{5}} + \cancel{\frac{1}{5}} - \cancel{\frac{1}{7}} \ldots. + \cancel{\frac{1}{n-2}} - \frac{1}{n}\right)$$
$$= \frac{1}{2} - \frac{1}{2n}$$
$$= \frac{n}{2n} - \frac{1}{2n}$$
$$= \frac{n-1}{2n}$$

(1 mark)

Question 6 (3 marks)

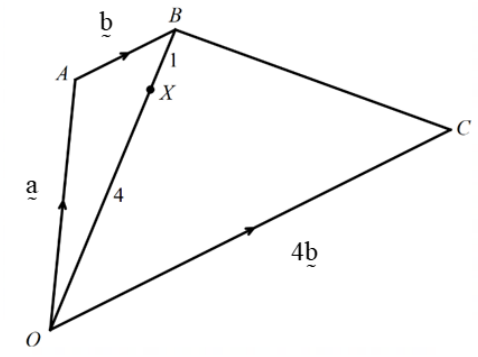
a. Draw a diagram to visualise the information given (see below).

To show that A, X and C are collinear, we need to show that there is a value m such that $\overrightarrow{AX} = m\overrightarrow{AC}$.

(1 mark)

$$\overrightarrow{AX} = -\underset{\sim}{a} + \frac{4}{5}\overrightarrow{OB}$$
$$= -\underset{\sim}{a} + \frac{4}{5}(\underset{\sim}{a} + \underset{\sim}{b})$$
$$= -\frac{1}{5}\underset{\sim}{a} + \frac{4}{5}\underset{\sim}{b}$$
$$\overrightarrow{AC} = -\underset{\sim}{a} + 4\underset{\sim}{b}$$
$$= 5\overrightarrow{AX}$$

$$\therefore \overrightarrow{AX} = \frac{1}{5}\overrightarrow{AC}$$



Therefore A, X and C are collinear. **(1 mark)**

b. Given $\overrightarrow{AC} = 3\overrightarrow{CY}$ and knowing $\overrightarrow{AC} = -\underset{\sim}{a} + 4\underset{\sim}{b}$,

$$3\overrightarrow{CY} = -\underset{\sim}{a} + 4\underset{\sim}{b}$$
$$\overrightarrow{CY} = \frac{1}{3}(-\underset{\sim}{a} + 4\underset{\sim}{b})$$
$$= -\frac{1}{3}\underset{\sim}{a} + \frac{4}{3}\underset{\sim}{b}$$

$$\therefore \overrightarrow{OY} = 4\underset{\sim}{b} + \left(-\frac{1}{3}\underset{\sim}{a} + \frac{4}{3}\underset{\sim}{b}\right)$$
$$= -\frac{1}{3}\underset{\sim}{a} + \frac{16}{3}\underset{\sim}{b}$$

(1 mark)

Question 7 (5 marks)

$$\frac{dy}{dx} = (2x+1)\sqrt{(9-y^2)(x-2)}$$
$$= (2x+1)\sqrt{x-2}\sqrt{9-y^2}$$

Using separation of variables,

$$\int \frac{1}{\sqrt{9-y^2}}\,dy = \int (2x+1)\sqrt{x-2}\,dx$$

(1 mark) – attempt to separate variables

$$\int \frac{1}{\sqrt{9-y^2}}\,dy = \sin^{-1}\left(\frac{y}{3}\right)$$

$\int (2x+1)\sqrt{x-2}\,dx$ is found using linear substitution:

Let $u = x-2$, $\therefore \frac{du}{dx} = 1$ and $x = u+2$

$$\therefore \int (2x+1)\sqrt{x-2}\,dx = \int (2(u+2)+1)\sqrt{u} \times \frac{du}{dx}\,dx$$
$$= \int (2u+5)\sqrt{u}\,du$$
$$= \int 2u^{\frac{3}{2}} + 5u^{\frac{1}{2}}\,du$$
$$= \frac{4}{5}u^{\frac{5}{2}} + \frac{10}{3}u^{\frac{3}{2}} + c$$

(1 mark) – attempt linear substitution

Putting both sides together and replacing u with $(x-2)$ give us

$$\sin^{-1}\left(\frac{y}{3}\right) = \frac{4}{5}(x-2)^{\frac{5}{2}} + \frac{10}{3}(x-2)^{\frac{3}{2}} + c$$

(1 mark) – both antiderivatives correct

Given $y(2) = 3$,

$$\sin^{-1}\left(\frac{3}{3}\right) = \frac{4}{5}(2-2)^{\frac{5}{2}} + \frac{10}{3}(2-2)^{\frac{3}{2}} + c$$
$$\frac{\pi}{2} = c$$

(1 mark) – attempt to find c

$$\therefore \sin^{-1}\left(\frac{y}{3}\right) = \frac{4}{5}(x-2)^{\frac{5}{2}} + \frac{10}{3}(x-2)^{\frac{3}{2}} + \frac{\pi}{2}$$

Solving for y:

$$\frac{y}{3} = \sin\left(\frac{4}{5}(x-2)^{\frac{5}{2}} + \frac{10}{3}(x-2)^{\frac{3}{2}} + \frac{\pi}{2}\right)$$
$$y = 3\sin\left(\frac{4}{5}(x-2)^{\frac{5}{2}} + \frac{10}{3}(x-2)^{\frac{3}{2}} + \frac{\pi}{2}\right)$$

(1 mark)

Question 8 (3 marks)

To find the stationary point(s), we need to use implicit differentiation.

$$2x - x\frac{dy}{dx} - y + 2y\frac{dy}{dx} = 0$$ **(1 mark)**

$$\frac{dy}{dx}(2y - x) = y - 2x$$
$$\frac{dy}{dx} = \frac{y-2x}{2y-x}$$

The stationary point occurs when $\frac{dy}{dx} = 0$, therefore $y - 2x = 0$

$$y = 2x$$ **(1 mark)**

Alternatively, you could substitute $\frac{dy}{dx} = 0$ into first line found above to get $2x - y = 0$, which leads to $y = 2x$.

Substituting $y = 2x$ into the original equation gives:

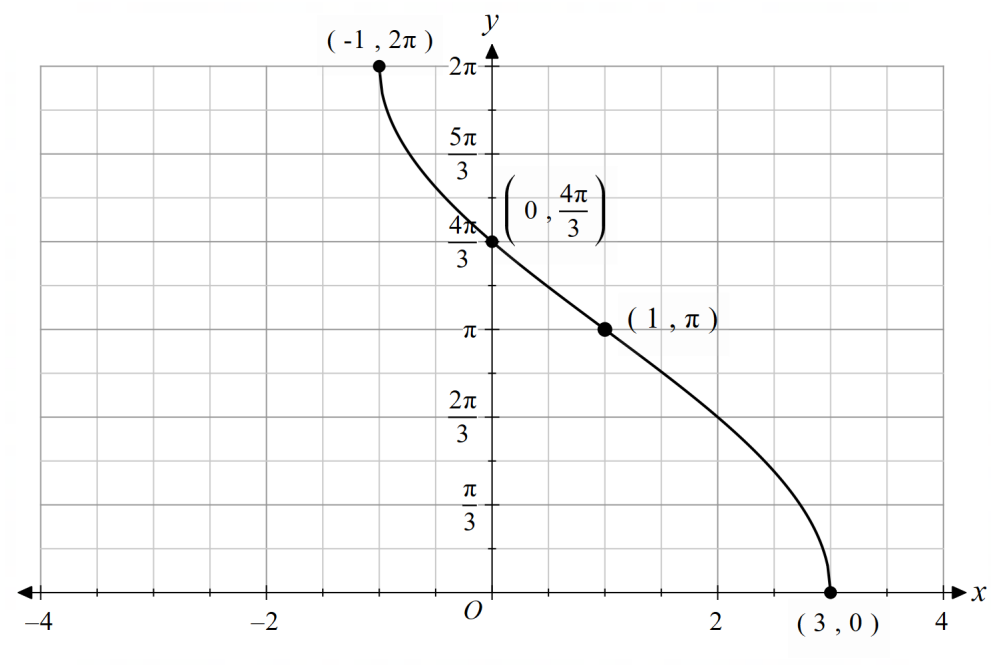
$$x^2 - x(2x) + (2x)^2 = 12$$
$$3x^2 = 12$$
$$x = \pm 2$$

When $x = 2, y = 4$ and when $x = -2, y = -4$.

The stationary points are at $(2, 4)$ and $(-2, -4)$. **(1 mark)**

Question 9 (6 marks)

a.



(1 mark) – endpoints in correct spot and labelled with coordinates
(1 mark) – y-intercept and point of inflection in correct spot and labelled with coordinates
(1 mark) – shape (approximate arccos shape including tangent being vertical at endpoints

b. When rotating a graph about the y-axis, we need to rearrange the rule to make x the subject.

$$y = 2\cos^{-1}\left(\frac{x-1}{2}\right)$$

$$\frac{y}{2} = \cos^{-1}\left(\frac{x-1}{2}\right)$$

$$\cos\left(\frac{y}{2}\right) = \frac{x-1}{2}$$

$$\therefore x = 2\cos\left(\frac{y}{2}\right) + 1$$

The volume of solid of revolution is therefore $V = \pi\int_0^{2\pi}\left(2\cos\left(\frac{y}{2}\right)+1\right)^2 dx$.

(1 mark) – attempt to rearrange and place into integral correctly

$$V = \pi\int_0^{2\pi}\left(2\cos\left(\frac{y}{2}\right)+1\right)^2\,dx$$

$$= \pi\int_0^{2\pi} 4\cos^2\left(\frac{y}{2}\right)+4\cos\left(\frac{y}{2}\right)+1\,dx$$

$$= \pi\int_0^{2\pi} 4\left(\frac{1}{2}(1+\cos y)\right)+4\cos\left(\frac{y}{2}\right)+1\,dx$$

(1 mark) – recognise use of double angle formula

$$= \pi\int_0^{2\pi} 2\cos(y)+4\cos\left(\frac{y}{2}\right)+3\,dx$$

$$= \pi\left[2\sin(y)+8\sin\left(\frac{y}{2}\right)+3y\right]_0^{2\pi}$$

$$= \pi\left[(0+0+6\pi)-(0+0+0)\right]$$

$$= 6\pi^2 \text{ cubic units}$$ **(1 mark)**

Question 10 (5 marks)

The integration by parts formula is given on the formula sheet and is as follows:

$$\int u\frac{dv}{dx}\,dx = uv - \int v\frac{du}{dx}\,dx$$

If we let

$u = e^{2x}$ and $\frac{dv}{dx} = \cos(3x)$

$\frac{du}{dx} = 2e^{2x}$ and $v = \frac{1}{3}\sin(3x)$

Using the integration by parts formula,

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} - \frac{2}{3}\int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx\,.$$

(1 mark) – recognise and attempt integration by parts

<u>Method 1</u>

Given that after performing integration by parts the first time, we still have an exponential and trigonometric function, we perform integration by parts on the integral $\int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx$:

If we let

$$u = e^{2x} \qquad \text{and} \qquad \frac{dv}{dx} = \sin(3x)$$

$$\frac{du}{dx} = 2e^{2x} \qquad \text{and} \qquad v = -\frac{1}{3}\cos(3x)$$

$$\therefore \quad \int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx = \left[-\frac{1}{3}e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} + \int_0^{\frac{\pi}{3}} \frac{2}{3}e^{2x}\cos(3x)\,dx$$

(1 mark) – use integration by parts again

Replace this integral back into the original integration by parts performed:

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} - \frac{2}{3}\left(\left[-\frac{1}{3}e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} + \int_0^{\frac{\pi}{3}} \frac{2}{3}e^{2x}\cos(3x)\,dx\right)$$

$$= \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} + \frac{2}{9}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} - \frac{4}{9}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx \qquad \textbf{(1 mark)}$$

$$\therefore \frac{13}{9}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} + \frac{2}{9}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}$$

$$\therefore \int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \frac{9}{13}\left(\left[\frac{1}{3}e^{2x}\sin(3x) + \frac{2}{9}\left(e^{2x}\cos(3x)\right)\right]_0^{\frac{\pi}{3}}\right) \qquad \textbf{(1 mark)}$$

$$= \frac{9}{13}\left(\left(0 + \frac{2}{9}e^{\frac{2\pi}{3}}(-1)\right) - \left(0 + \frac{2}{9}(1)\right)\right)$$

$$= \frac{9}{13}\left(-\frac{2}{9}e^{\frac{2\pi}{3}} - \frac{2}{9}\right)$$

$$= -\frac{2}{13}\left(e^{\frac{2\pi}{3}} + 1\right) \qquad \textbf{(1 mark)}$$

<u>Method 2</u>

Having found $\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} - \frac{2}{3}\int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx$ ….. equation (1)

Note that this still gives us an integral with an exponential and trigonometric function in it.

We can use integration by parts again and solving the two equations simultaneously.

If we let

$u = \cos(3x)$ and $\frac{dv}{dx} = e^{2x}$

$\frac{du}{dx} = -3\sin(3x)$ and $v = \frac{1}{2}e^{2x}$

$\therefore \int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{2}e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} - \int_0^{\frac{\pi}{3}} -\frac{3}{2}e^{2x}\sin(3x)\,dx$ ….. equation (2)

(1 mark) – recognise to use integration by parts again

Re-arranging equation (2)...

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx - \left[\frac{1}{2}e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} = \frac{3}{2}\int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx$$

$$\frac{2}{3}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx - \frac{2}{3}\left[\frac{1}{2}e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} = \int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx$$

$$\frac{2}{3}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx - \frac{1}{3}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}} = \int_0^{\frac{\pi}{3}} e^{2x}\sin(3x)\,dx$$

(1 mark) – attempt to rearrange one of the equations and perform substitution

Substitute this into equation (1)…..

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} - \frac{2}{3}\left[\frac{2}{3}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx - \frac{1}{3}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}\right]$$

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} - \frac{4}{9}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx + \frac{2}{9}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}$$

$$\frac{13}{9}\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} + \frac{2}{9}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}$$ **(1 mark)**

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \frac{9}{13}\left[\frac{1}{3}e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} + \frac{2}{13}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}$$

$$\int_0^{\frac{\pi}{3}} e^{2x}\cos(3x)\,dx = \frac{3}{13}\left[e^{2x}\sin(3x)\right]_0^{\frac{\pi}{3}} + \frac{2}{13}\left[e^{2x}\cos(3x)\right]_0^{\frac{\pi}{3}}$$

$$= 0 + \left[\left(\frac{2}{13}e^{\frac{2\pi}{3}}\cos(\pi)\right) - \left(\frac{2}{13}e^{0}\cos(0)\right)\right]$$

$$= -\frac{2}{13}e^{\frac{2\pi}{3}} - \frac{2}{13}$$ **(1 mark)**

$$= -\frac{2}{13}\left(e^{\frac{2\pi}{3}} + 1\right)$$